RegBridge: Inspectable AI-Assisted Decisions for Regulatory Document Reuse

Ashraf Abdul-Muumin[1]*, Shahram Rahimi[1], Raj Krishnan[2], and Kaushik Roy[1]

[1]Department of Computer Science, University of Alabama, Tuscaloosa, Alabama, USA
[2]AIFP
aashraf@crimson.ua.edu, srahimi1@ua.edu, kroy2@ua.edu, Raj@aiforservices.com

Abstract

A technically referenceable regulatory document can still require a new placement, contextual checks, or human review. RegBridge is an interactive research prototype for inspecting these distinctions in prospective FDA/CDER electronic Common Technical Document (eCTD) reuse. Users upload a synthetic dossier, confirm its target context, and inspect document-level decisions, proposed actions, source evidence, and an explanation graph. Deterministic parsing and scoped rules handle exact structural facts; a language model supplies citation-bounded semantic findings, not final dispositions. The demonstration contrasts a removed heading, lifecycle-sensitive metadata, and stale applicant wording, and distinguishes incomplete semantic inspection from execution failure. A separate frozen experiment on 12 author-adjudicated held-out cases produced 11 correct decisions in each of three executions, with no observed unconditional reuse of eight action-required cases. These are controlled observations, not regulatory validation or a general safety guarantee.

Why Demonstrate Regulatory Reuse?

Reusing a document is not the same as preserving its regulatory context. A legacy eCTD v3.2.2 document may remain identifiable while its heading disappears from a selected v4.0 hierarchy, its metadata requires an explicit lifecycle choice, or its text names a previous applicant. A useful assistant must show *which fact changes the action*, not merely generate a plausible explanation.

RegBridge targets prospective FDA/CDER decision support, not conversion or acceptance prediction. FDA accepts new v4.0 applications but describes forward compatibility for existing v3.2.2 applications as a future phase (U.S. Food and Drug Administration 2024). The interface therefore labels this scenario `not_operational`. It does not rewrite regulated documents, create submission-ready packages, or certify compliance.

The contribution is an inspectable integration of bounded semantic analysis with source-versioned lifecycle rules and an explicit human-review boundary. Related work already combines retrieval, validation, and human oversight: ORCHID addresses policy-grounded property classification (Mahbub et al. 2025), while Condition Insight constrains industrial-maintenance reasoning with deterministic evidence and rule-based verification (O'Donncha et al. 2026). RegBridge specializes this pattern to document identity, context placement, and lifecycle intent; it does not claim to invent hybrid reasoning.

*Corresponding author.

What the Attendee Does

The React/FastAPI prototype offers an upload Analyzer, package-derived Baselines, guided cases, and a separate frozen Evaluation view. The attendee loads the bundled public synthetic dossier or uploads a supported ZIP, confirms FDA/CDER scope and identifier-based reuse, chooses lifecycle intent, and runs analysis. Both XML backbones are checked offline against pinned ICH/FDA DTDs. Profile acceptance and per-document policy coverage are displayed separately: valid XML does not establish reuse eligibility. Archive-provided DTDs are never executed.

The three-document walkthrough makes the reasoning boundary concrete. **A: placement.** A document under `3.2.S.1.2` activates the exact scoped mapping to `3.2.S.1`; the attendee opens its rule, source locator, and proposed new-context action with legacy-context suspension (U.S. Food and Drug Administration 2025, 2026). This is a proposed action, not an executed conversion. **B: intent.** For `manufacturer="all"`, explicit preservation retains the existing context with an advisory; unspecified intent requests review (U.S. Food and Drug Administration 2017). The upload UI deliberately leaves manufacturer partitioning unknown, so selecting normalization requests that missing information rather than inventing it. **C: meaning.** Stale applicant wording in a PDF is compared with parsed regional metadata. The attendee opens the cited occurrence and sees why a structurally acceptable document still needs human review.

Opening a result reveals its action, evidence, actual adapter, execution status, trace, and graph. Baselines reuses the inventory without benchmark labels or accuracy claims. Evaluation displays historical results, not a recomputation on the upload. Fixture mode is explicitly disclosed and needs no model key; the live profile uses GPT-5.5 through a structured-output adapter. The local-model option is disabled.

From Findings to Inspectable Decisions

Figure 1 presents the implementation flow. The parser produces document, metadata, and structural evidence occur-

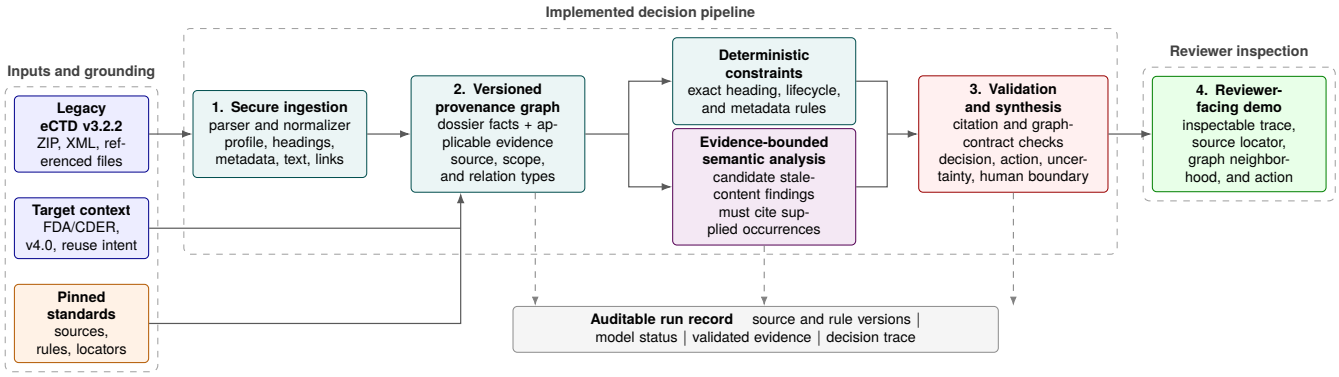


Figure 1: Implementation flow of RegBridge. Deterministic ingestion, versioned provenance, and executable constraints establish the decision boundary. The language model contributes only candidate semantic findings grounded in supplied evidence; validation and deterministic synthesis govern the published action and human-review requirement.

rences with owners and locators. Versioned rule records retain source digests, applicability, verification basis, and review status. Only author-adjudicated direct encodings or mechanical derivations may be hard rules; interpretations remain advisory and model findings remain semantic signals. None is regulatory-expert validated.

For request evidence E, a publishable finding must satisfy its schema, cite only occurrences in E, and satisfy the graph's typed relation contract. A finding `CITES` an evidence occurrence, not a normalized keyword; an `ABOUT` keyword must match the keyword `OBSERVES` from that occurrence. This separates mention identity from concept identity, complementing ontology-aware extraction (Chepurova et al. 2026). These checks establish provenance consistency, not semantic entailment. The explanation graph is constructed and validated before publishing the synthesized result; graph search does not select the decision.

For fixed parsed facts, rules, and validated model output, synthesis is deterministic. In the demonstrated Case A, semantic abstention preserves the structural action but marks inspection incomplete and requires human approval. In B/C, abstention requests completion of inspection without fabricating stale-content evidence. A transport, schema, graph, or persistence failure produces a separate failure record, not a regulatory decision. These branches are software contracts, not calibrated risk estimates or a proof of completeness across overlapping rules.

Evidence and Limits

Frozen experiment. The original 30-case benchmark has 12 train, six development, and 12 held-out cases across six held-out families. The test set has four references each for legacy reuse, new-context reuse, and human review; the six permitted output labels were available to all systems. B0 receives all six frozen standards spans, B1 retrieves BM25 top-3, B2 omits model semantics, and RegBridge combines rules and bounded semantic analysis. GPT-5.5 (provider-reported `2026-04-23`) used medium reasoning, 4,000 maximum output tokens, and an 800-token final-answer limit; temperature was unsupported. Three executions yielded 108 outcomes: 105 model-backed and three deterministic branches, without voting. B2 was scored separately once.

Table 1: Frozen held-out observations; ranges span three executions of the same 12 cases. U: unsafe reuse (/8); B: review bypass (/4); C: clean unconditional reuse (/4).

System	Accuracy	Macro-F1	U	B	C
B0	.417–.583	.504–.641	0	1	0–1
B1	.417–.500	.448–.556	0	0–1	0
B2	.833	.822	2	2	4
RegBridge	.917	.915	0	0	4

An action-required reference specifies a non-legacy decision, material action, or human review. *Unsafe reuse* means predicting unconditional legacy reuse for such a case. *Review bypass* means a human-review reference receives neither that decision nor a human-review flag. B2 bypassed two applicant-mismatch cases. RegBridge instead over-deferred A005 to review rather than new-context reuse. Its decisions agreed across repetitions on 12/12 cases; abstention state changed on one of 11 model-invoked cases. Token-usage variation alone is not semantic variation.

B0/B1 emitted unrepresented labels on 3–4/12 and 3/12 cases, respectively, confounding exact-match comparisons. A descriptive binary projection (unconditional reuse versus any required intervention) yields 8–9/12 correct for B0, 8/12 for B1, 10/12 for B2, and 12/12 for RegBridge. This retains all outputs but does not measure repair correctness or remove architectural confounding. Clean-reuse counts prevent interpreting blanket intervention as useful safety.

Current prototype. The three-case and failure-handling paths pass 55 targeted network-free integration tests. No regulatory professional validated the author-defined labels. Only three output classes are represented, and rule-composition coverage is limited.

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